UME630 : Total Quality Management

L	T	P	Cr
3	0	0	3.0

Course Objective: This course provides students with the knowledge required to assess and improve product quality through process control procedures and quality improvement techniques. The course also provides required knowledge for line/angular measurements.

Syllabus

Quality Management: Meaning and significance of 'Quality'; Fitness levels of quality: characteristic features and limitations of each type; Essential components of Quality; Product Features; Freedom from Deficiencies; Characteristics and attributes under each for products and services; Phases or elements of building quality in a product: Quality of design, Quality of conformance, Quality of performance, History of quality; Evolution of the concepts of quality; Big Q Vs Small Q: changing scope of quality activities; Ishikawa's seven quality tools; Quality circles; TQM meaning and TQM culture.

Quality System Economics: Different obvious quality costs; Hidden quality costs; Economic models of quality costs: Conventional and modern models; cost curves: Different zones and actions to be taken for each zone.

Quality Control and Improvement: Causes of variation: special and chance causes; sporadic and chronic problems; Juran's Quality Trilogy; Different Zones and their functions; Quality Planning activities; Crosby's and Deming's quality viewpoint; Variable Vs Attribute Data; Different distributions, their significance and applications; Control charts for variables; underlying principle; advantages; limitations and applications of: X-bar & R charts, X-bar & S charts, X & MR charts. Control charts for attributes; underlying principle; advantages, limitations and applications of: p, np, c and u charts.

Process Capability Analysis: Need and significance; Process capability for variable data; Process capability indices; interpreting the indices; Six Sigma Process Quality.

Tools and Techniques for Quality Management: Quality functions development (QFD); Failure mode and effect analysis (FMEA); Seven new management tools; Poka Yoke; Total preventive maintenance (TPM); Implementing TQM: an integrated system's approach.

Experimental Design and the Taguchi Method: Experimental design fundamentals; factorial experiments; Taguchi method.

Metrology: Introduction to metrology; methods of measurement; measurement standards; calibration, linear metrology; measurement of surface finish; measurement of flatness and cylindricity; use of comparators; screw thread metrology.